

LLM-based Quiz Generation for Assessments in Learning Management System

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Abstract— An advanced AI system which employs Large Language Models (LLMs) to create automatic assessment questions through automation features, has been introduced in this study. The system combines MetaAI's llama-3 model Groq API together with LangChain and structured prompting methods to produce quizzes which remain accurate to original contexts and logical sequences. The system delivers better efficiency alongside reduced manual work through the creation of quality questions across different formats which contain multiple-choice items, true-false formats and open-ended questions. The system maintains data security through environment variables in its API handling process to protect against unauthorized access. Smart testing through AI-based quiz creation methods produces customized assessments which use dynamic test content according to the established study.

Keywords— Large language models, llama-3, Structured prompting, AI-driven quiz generation

I. INTRODUCTION

The development of Learning Management Systems (LMS) under educational digitalization brought about new educational patterns by focusing on authentication security and enabling course administration systems alongside dynamic assessment tools. Educational establishments together with independent learners require adaptable LMS platforms with adjustable characteristics and multiform learning abilities since they both transfer educational delivery to digital networks. Current LMS tools employ static testing methods to conduct knowledge evaluations but do not improve results. Standardized educational content is produced for all students through assessment approaches that have not changed. The limited capabilities of evaluation feedback systems hinder teachers from providing personalized assessment feedback that would help students learn better thus reducing student motivation levels.

Teaching staff measure student content understanding through the educational measurement framework by using testing activities to track student development. Manually-guided question creation by teachers amounts to resource depletion and prolonged development procedures for quizzes that produce standardized questions which cannot be modified. The generation of quizzes through automated methods by AI depends on Large Language Models to produce accurate educational content written with proper grammar. The platform generates adaptive quiz assessments with assistance from a Large Language Model (LLM) to extend beyond conventional LMS functionality. Student performance data processed by the system enables platform operation to generate personalized learning opportunities that fit each individual student.

The adaptive testing system generates tailored questions for students to work on at their present competency level which first targets weak areas then progresses learning at their current level. Students encounter tailored examination questions within the system and receive additional details on student activity metrics and test outcomes and course wide progress tracking data for instructors. Educational methods created through information collection processes enable teachers to improve student learning outcomes effectively. The LMS accomplishes innovative classification by utilizing Next.js together with LLM-driven testing to improve educational functions.

The purpose of the research involves developing an AI-based quiz generation system which provides scalable features to help educators produce questions efficiently. The system uses modern AI technology to automate quiz development which brings substantial reductions to educator work without compromising question quality. Research shows that artificial intelligence brings advantages to automated assessment systems by incorporating practical quiz generation and it will shape future development of efficient assessment solutions operating at scale. The suggested LMS functions to merge between conventional online education and individualized learning models which rely on data.

II. RELATED WORKS

Anil et al. [1] researched PaLM 2, which marks the beginning of a fresh state-of-the-art language model generation because it improves multilingual understanding while enhancing both operational efficiency and reasoning performance. The Transformer model PaLM 2 delivers superior performance than its predecessor by merging training objectives that generate substantial improvements across BIG-Bench reasoning and question-answering tests in multiple languages along with programming capabilities and translation features. The current research shows that PaLM 2 achieves efficient data scaling methods alongside optimized data preparation techniques and implements real-time control functions to create responsible AI operations. Independent experts support the superiority of PaLM 2 over previous models due to its efficient cost-effective capabilities which creates opportunities for future practical implementations.

Rao et al. [2] executed an extensive evaluation of automatic MCQ generation approaches with special emphasis on their applications in standardized assessments. The experts designed a workflow system which contained six steps beginning with pre-processing and ending with post-processing. The research presented different techniques for each phase while recognizing the necessity to develop standardized evaluation approaches. Future researchers need to create standardized evaluation metrics and benchmark

datasets to enhance the performance of MCQ generation systems according to the authors' conclusion.

Mehta et al. [3] presented a system that utilizes NLP techniques to produce MCQs from text through an automated process. The work describes the system design starting from text analytics which leads to concept retrieval for making questions and distractors. The planned automation of MCQ production through this approach works to both boost efficiency and cut down content delivery time while maintaining relevant questions that match the educational subject matter.

Nema et al. [4] evaluated the problems of conventional n-gram-based metrics such as BLEU and METEOR for measuring automatically created questions. The authors developed a fresh scoring model for answerability evaluation which displayed better agreement with human scores when used alongside present evaluation approaches. The research shows that evaluation method refinement leads to major improvements in the dependability of automated question generation systems.

Nwafor et al. [5] developed a keyword extraction system that converts learning content into multiple-choice questions (MCQs). The automatic question generation process of the system, minimizes teachers' workload by facilitating simple generation of questions for examination. The system extracted examinable content effectively by generating results that matched keywords chosen by instructors under evaluation carried out on five learning materials. This research offers an effective method that makes examination preparation easier at a large scale.

Raina et al. [6] proposed a system that could facilitate machine-automatic generation of Multiple-Choice Question (MCQ) development for educational testing. Raina et al. developed quality questions and answers from natural language processing tools that worked on input text. The authors emphasized the significance of relevant, accurate, and diverse content in MCQs generated automatically as part of the contribution in designing scalable automated testing processes. The contribution introduced significant improvement in MCQ development by preventing limitations with regard to efficient and fair generation of MCQs.

Tan et al. [7] developed a chatbot platform to merge Peer Instruction and Just-in-Time Teaching (JiTT) flipped education methods. LLMs served as the foundation for a new classroom flipping system which produced peer questions by AI algorithms to increase student participation. Student involvement increased and instructors could adapt their teaching pace because LLMs produced questions that promoted student-oriented learning according to research findings. The authors also discussed the potential for AI-generated false information, so instructor oversight was introduced as a means to preserve educational credibility.

Chernov et al. [8] examined Mixture of Experts (MoE) layers within LLMs for their performance in quiz-based tasks across the MMLU benchmark. The analysis of OLMoE model showed that experts remained inactive during inference operations in more than 60% of cases therefore indicating potential efficiency improvements by removing these unused experts. The uniform distribution from the gating network produced challenges for model robustness because it differed from natural sparse patterns. Better performance and reduced

computational demands emerge when the model selects experts with high accuracy rates according to the research.

Zhang et al. [9] examined if synthetic educational data produced by LLMs could function as an assessment method for unstructured educational queries. A research study consisting of two phases showed synthetic data delivered excellent training results under controlled conditions yet did not perform well when implemented in standard practice. The research suggested using synthesized and actual educational data in unison to enhance AI-assisted educational assessment technologies.

Fan et al. [10] discussed the application of Large Language Models (LLMs) in the classroom environment through the Educational LLM Framework (ELF) to enhance the quality and efficacy of AI-generated content in pedagogy. The study discussed the boundedness and inaccuracy issues of LLMs in the learning environment and introduce a holistic framework that combines sophisticated prompting mechanisms and evaluation metrics to generate high-quality dialogical content and structured pedagogical materials. The ELF seeks to enhance engagement and pedagogical style through context-relevant dialogue completion and expertise transfer learning conditions. The method involves a blend of Perplexity-based Statistical Evaluation and Human Evaluation with Questionnaires, seeking to generate content that is semantically unambiguous and pedagogically accurate. The article introduces a new development in the application of AI in the learning and teaching process, recognizing the potential of LLMs to transform traditional pedagogical practices.

Perez et al. [11] examined how Large Language Models (LLMs) working with Retrieval-Augmented Generation (RAG) helped deliver education at the graduate level of biological chemistry. The AI system ChemTask operated through an AI design that helped students receive academic responses to the queries. The evaluation confirmed that ChemTask matched human teaching assistants in performance while demonstrating strength in problems that demanded creative solutions. Research shows that binding Large Language Models with Retrieval-Augmented Generation produces better academic support platforms which enhance student learning achievement.

Noorbakhsh et al. [12] of the M.I.T. Computer Science and Artificial Intelligence Lab presented 'Savaal,' an AI-based scalable question generation system that aims to enhance human learning by automating depth-of-understanding questions and encouraging critical thinking. Unlike traditional systems with a focus on memorization of facts, Savaal is created to create questions that encourage learners to engage in conceptual thinking and synthesis in a wide range of topics. The system goes through a three-stage pipeline, including concept extraction, passage retrieval related to the concept, and question generation with the application of large language models (LLMs). The process not only scales well over large collections of documents but also ensures that generated questions are contextually rich and high in educational value. The research presents a comprehensive human evaluation, showing that Savaal outperforms traditional methods by far in generating questions that are stimulating and useful in facilitating higher-level educational achievements.

Wang et al. [13] introduced 'Generative Students', a novel system based on simulated student profiles using LLM that aid in evaluating question items in education. The system is based

on the Knowledge-Learning-Instruction (KLI) model, where knowledge components are discovered to be acquired by students. Dynamically generated profiles with varying levels of comprehension and confusion among students are generated and employed to generate and test MCQs. The authors give a discussion of how the structure facilitates the auto-generation of a variety of different student profiles and how it facilitates the generation of MCQs that match simulated states of knowledge in students. The research illustrates that generative student responses can simulate responses of real students, offering a novel mode of assessment of question items and improving the quality of educational tests.

Zeinalipour et al. [14] The system uses Large Language Models (LLMs) to create Turkish educational quizzes through automation. The Turkish-Quiz-Instruct dataset developed contains Turkish educational texts together with multiple choice questions. After fine-tuning GPT-3.5-Turbo, Llama-2-7b-chat-hf alongside Llama-2-13b-chat-hf all led to substantial enhancements in the quality of questions generated. The research demonstrated that LLMs are useful in educational technology beyond English while establishing basic principles which will advance low-resource language question generation technology in the future.

Hasan et al. [15] developed a question-and-answer generation framework which employs a trained LLM to produce various questions types such as multiple-choice alongside conceptual questions. The system operates with Meta-Llama 2-7B architecture and receives training through the RACE dataset while utilizing prompt engineering to create various questions for differing difficulty ranges. The system functions to automate education assessment by cutting manual work while creating an unbiased question database with varied content. According to the research LLMs demonstrate the possibility to automate academic assessments by retaining question meaning alongside logical connection.

Grévisse et al. [16] investigated docimological quality in MCQs generated by LLMs for computer science and medicine courses. The study investigates how far the GPT-based systems generate questions according to best MCQ writing practices by using a Moodle plugin to automate test creation. The study demonstrates LLM competency in generating MCQs even though some examples include low-quality or unclear distractors that need human editing to uphold instructional quality.

Hwang et al. [17] examined AI-assisted methods for generating and evaluating multiple-choice questions (MCQs) in first-year chemistry and biology. The approach employed GPT-3.5 in zero-shot prompting to produce questions aligned with Bloom's Taxonomy. RoBERTa was employed to confirm the taxonomy alignment and evaluate the quality of questions against Item Writing Flaws (IWF) standards. The research concludes that GPT-3.5 can produce higher-order thinking questions but that there are occasional differences between AI-produced questions and those judged by human experts. The authors mention the implication of the findings for the use of AI in educational testing, i.e., while AI can greatly enhance the range and efficiency of question generation, great care must be exercised in the use of human judgment to determine the pedagogical validity of the questions.

Hang et al. [18] described MCQGen as an educational framework based on AI that produces automated multiple-choice questions. The framework unifies a large language

model with retrieval-augmented generation along with superior prompt engineering to automatically create relevant and precise MCQs. MCQGen provides customization of student learning through adaptive question complexity and specialized correction of frequent student misunderstandings. Through automated MCQ generation students complete their assignments at half the usual time expenditure to acquire higher levels of learning achievement in blended or flipped classrooms.

Bhowmick et al. [19] introduced the design and assessment of an automated question generation system for educational assessment. The work introduced the widespread application of question-based activity in education, traditionally a core aspect of learning and testing. Recent research developments in generative AI, i.e., transformer-based language models, to automate the generation of multiple-choice questions (MCQs) from text material, was used. The system is designed with separate modules to assist in question generation, correct answer prediction, and distractor creation. The modularity allows for the possibility of developing and refining individual modules in isolation to align with the strengths and limitations of different generative models. The comprehensive qualitative and quantitative assessment offers trade-offs in the use of different techniques and models, with insightful indications of the potential of automated question generation to reduce teacher workload and tailor learning experiences.

Scaria et al. [20] investigated the ability of large language models to generate education questions of various cognitive levels based on Bloom's taxonomy. The study assessed the performance of state-of-the-art LLMs of varying sizes employing state-of-the-art prompting techniques to determine the ability of the models to generate diverse and high-quality education questions. The authors conduct both expert and LLM-based evaluation to determine the linguistic and pedagogical quality of the generated questions. The findings indicate that there are significant differences in the performance of the models, where some LLMs possess the ability to generate questions that have the potential to tackle higher cognitive capabilities. The research determines the ability and limitations of using LLMs in automated education question generation, which requires the further development of prompting techniques and evaluation measures in order to achieve the full capability of AI in education.

Llama Team [21] introduced a detailed description of Llama 3, a new family of foundation models with the capability of multilinguality, coding, reasoning, and tool use. The largest model, a dense Transformer with 405B parameters, is intended to perform a wide range of tasks in many languages and formats. The study introduces the development and large-scale empirical evaluation of the models, compared with state-of-the-art language models such as GPT-4. The authors emphasize the new aspects of Llama 3, including its capability of combining image, video, and speech capabilities, and the experiments with them compared with state-of-the-art models in the respective areas. The open-source release of Llama 3 under a community license is aimed at enabling further research and development in AI, with particular emphasis on safety and ethical concerns in model deployment.

III. PROPOSED SYSTEM

An AI-based quiz generator produces assessment questions through the implementation of Large Language Models

(LLMs) as shown in Fig 1. The system manages a specific workflow order that starts with user course completion followed by assessment participation then moves to question generation from user input. User input enters the LLM system which generates quizzes that exist in various formats. High-quality contextually relevant questions emerge because the system combines structured prompts with secure API protections when generating questions. The system features the functionalities described for its individual components in the Figure below.

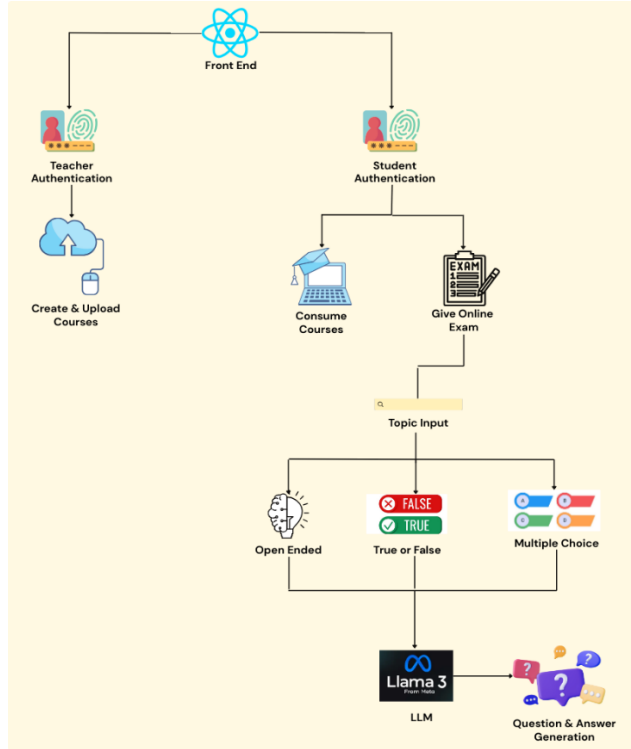


Fig. 1. Proposed system

User Experience

Teachers access an intuitive dashboard through the platform to create manage and upload course content with total ease. Teachers have the capability to structure lessons and enhance lessons with supporting PDFs and video links.

Students easily navigate their own individualized interface for enrolling in courses, accessing course materials, and monitoring their academic progress. A web-based testing system enables users to take tests online upon the completion of courses by selecting designated subjects with open-ended questions, multiple-choice questions, and true-false questions.

The LLM methodology process is divided into four steps:

A. User Input:

The process of creating quizzes is triggered when the course is completed, and users enter mandatory required information in terms of subject interests along with difficulty levels and question format selection. The system applies this input information and accordingly customizes the quiz according to the user requirement.

B. Topic Submission and Prompt Processing:

Structured queries are the outcome of precise topic needs to control the operations of the LLM system. In the context of question generation as a task, the structured method ensures that generated questions are appropriately contextually relevant, grammatically correct, and follow stipulated learning objectives.

C. LLM-Based Processing and Quiz Generation:

The system uses the API provided by Groq together with LangChain to transform structured prompts into multiple-question types including multiple-choice alongside true-false questions and open-ended questions. The LLM creates dynamic questions through its formulation process to maintain diversity along with coherence as it eliminates repetitive content.

D. Question and Answers:

The assessment questions created get stored and structured into an assessment format. The system establishes that answer options must be precisely written while retaining logical organization and free of ambiguity. The system provides users with the opportunity to review and improve the accuracy and clarity of questions before their presentation to end-users.

IV. IMPLEMENTATION

Fig. 2. Interface

Figure 2 showcases the user interface of the system. It represents the starting point of the quiz generation.

A. User Input:

Fig. 3. User Input

Figure 3 highlights the text input field where users enter the concept or topic for which a quiz has to be generated. For example, in the image, “Python” has been entered as the topic.

B. Enter Number of questions:

Fig. 4. Select number of questions

Figure 4 demonstrates the dropdown menu where users specify the number of questions required in the quiz.

C. Select the quiz Type:

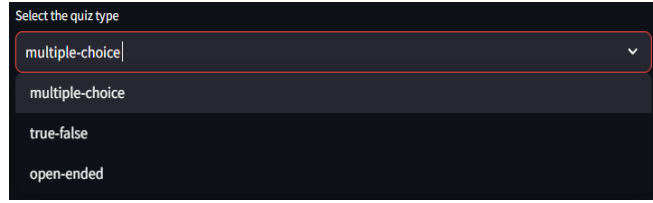


Fig. 5. Select quiz type

Figure 5 presents the different quiz formats available, including multiple-choice, true/false, and open-ended questions.

D. Question and Answers:

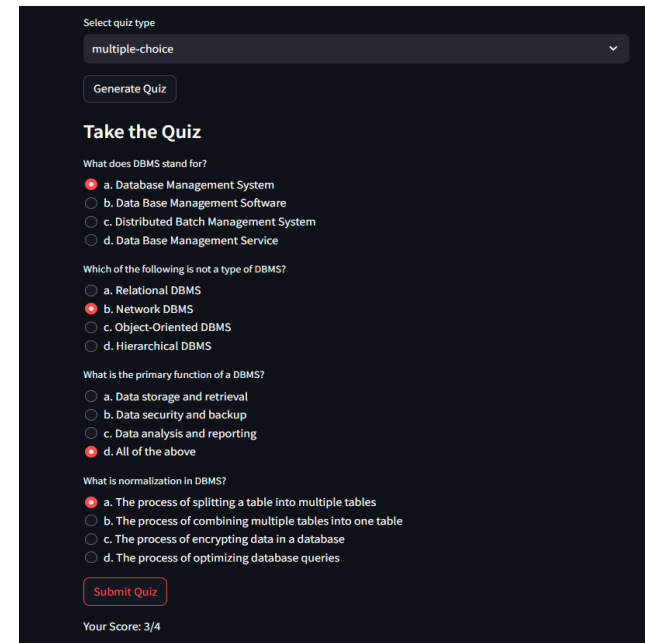


Fig. 6. Final output

Figure 6 displays the generated quiz, including multiple-choice questions, answer options, and the scoring system. The user interacts with the quiz and submits their responses, receiving an evaluation at the end.

V. RESULTS AND DISCUSSION

The evaluation of each model took place through assessments regarding inference time and resource consumption together with output quality evaluation. The research summary appears in the following table.

Table 1: Performance Table

Model	Inference Time (min)	RAM Usage (GB)	GPU Utilization	Output Quality
Mistral 7B	10 -15	16-18	~95%	Well-structured, sometimes oversimplified

Gemma	8-12	12-14	60-80%	Coherent but less detailed than Mistral
GPT 4o-mini	0.5-0.7 (30-40 sec)	~1.5	Minimal (cloud based)	High-quality, nuanced, insightful responses
LLaMa	4-5	7-9	~60%	High-quality, Detailed answers

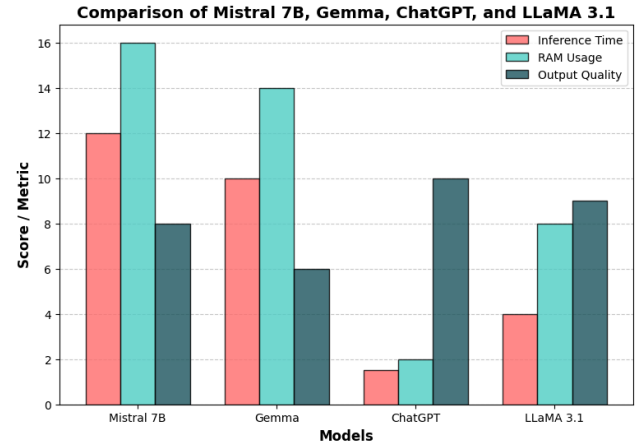


Fig. 7. Performance comparison graph

The performance metrics Inference Time, RAM Usage and Output Quality of Mistral 7B, Gemma, ChatGPT and LLaMA 3.1 are analyzed comparatively in table 1. An evaluation took place to measure the effectiveness along with efficiency of the different models.

The inference time performance metrics in Figure 8 indicate that LLaMA 3.1 possesses quicker speeds that make it the more appropriate model choice for time-sensitive applications compared to Mistral 7B and Gemma. The shorter inference operation time makes it more efficient for applications that use real-time assessment. ChatGPT is the quickest to process requests among the models tested but is not flexible and customizable and thus hampers its application for certain specialized operations.

From Figure 7, Mistral 7B and Gemma need more memory capacity to be efficient in their computation processes. Resource-limited environments are constrained by the high RAM requirements. LLaMA 3.1 strikes a balance between computation efficiency and efficient memory usage which is the best model deployment option for resource-limited environments. Learning management systems are enhanced by LLaMA 3.1 because of its balancing between RAM management and high productivity to provide efficient scalable solutions.

The research validates LLaMA 3.1 as a fast and resource-efficient model that produces top-notch outputs. The system's capability to produce quality results through economical computing makes it appropriate for practical uses such as the quiz generator in this study.

Llama 3 implements a standard dense Transformer model design according to Vaswani et al. (2017). The model design

of Llama 3 maintains alignment with Llama and Llama 2 (Touvron et al., 2023 a,b) because it features standard Transformer foundations.

Table 2. Overview of the key hyperparameters of Llama 3

	8B	70B	405B
Layers	32	80	126
Model Dimension	4,096	8192	16,384
FFN Dimension	14,336	28,672	53,248
Attention Heads	32	64	128
Key/Value Heads	8	8	8
Peak Learning Rate	3×10^{-4}	1.5×10^{-4}	8×10^{-5}
Activation Function	SWiGLU		
Vocabulary Size	128,000		
Positional Embeddings	RoPE($\theta = 500,000$)		

Table 2 presents a tabular overview of the key hyperparameters of the LLaMA 3 model across different versions (8B, 70B, 405B). It details specifications such as model dimension, Feedforward Network (FFN) size, attention heads, key/value heads, learning rate, activation function, vocabulary size, and positional embeddings.

VI. CONCLUSION

The system unites Large Language Models (LLMs) with quiz generation technology to develop automated assessment methods, which decreases the time required for creating exam questions through human intervention. The proposed system connects to the Groq API network that delivers advanced LLMs and integrates the LangChain framework to create prompt structures and process information. The model depends on the llama3-8b-8192 implementation because it demonstrates outstanding natural language processing abilities to create quiz questions while conserving context validity and maintaining flow patterns coupled with excellence in vocabulary. The system produces various high-quality questions using structured prompts which maintain logical order. The study demonstrates that LLM-based quiz generation provides superior benefits than traditional hand-made quiz approaches. The llama3-8b-8192 model functions as an essential tool for building contextual question creation in real time that can produce different assessment types including multiple-choice and open-ended responses. User inputs are subject to a structured workflow before the LLM receives the material for question creation to ensure educational context compatibility. The early system tests reveal that it produces tests with a good mix of high-quality structure and limited manual oversight, thereby allowing efficient large-scale production of quizzes. The system must be enhanced with improved techniques that support prompt engineering techniques to generate more precise specific questions. System reliability would be increased by incorporating validity tools to assess content validity, and the inclusion of support for measurement techniques unique to a domain. Adaptiveness is the most important system enhancement because the system would be capable of automatically adjusting question difficulty levels based on users' performance scores. Students can be given quizzes through adaptive test capabilities because the system can detect pace in an attempt to provide more advanced ideas and more challenging questions.

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